Project: Worked examples
Title: Column base plate (EN 1993-1-8)
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1. Scope and method

Calculates the bearing, bolt tension, T-stub and shear utilisations of a base plate under an I/H, RHS or CHS column on grout, and the plate thickness that satisfies them. The anchorage is designed outside this sheet.

WHAT IT CALCULATES	RELATION
Bearing (cl. 6.2.5)	$f_{cd} = \alpha_{cc}\cdot f_{ck}/\gamma_c$; $f_{jd} = \beta_j\cdot\alpha\cdot f_{cd}$, $\beta_j = 2/3$; $c = t_p\cdot\sqrt{[f_y/(3\cdot f_{jd}\cdot\gamma_{M0})]}$
Axial (cl. 6.2.8.2)	$N_{j,Rd} = f_{jd}\cdot A_{eff}$, A_{eff} the bearing strips of width c round the column footprint (full rectangle when they overlap); $u_N = N_{c,Ed}/N_{j,Rd}$
Moment (cl. 6.2.8.3, Table 6.7)	z_C from the section ((h − t_f)/2, (h − t)/2, half-annulus centroid) or entered, x_b the bolt row from the axis, $z = z_C + x_b$; $C_{Ed} = (M_{Ed} + N_{Ed}\cdot x_b)/z$ and $T_{Ed} = C_{Ed} − N_{Ed}$ when both are ≥ 0; with $N_{Ed} > 0$ and $T_{Ed} < 0$ both sides are in compression, $C = N_{Ed}/2 \pm M_{Ed}/(2z_C)$; with $N_{Ed} < 0$ and $C_{Ed} < 0$ both rows are in tension, $T = −N_{Ed}/2 \pm M_{Ed}/(2x_b)$; $F_{t,Ed} = T_{Ed}/n_t$ at the more loaded ro…
Plate T-stub (cl. 6.2.6.11, 6.2.4, Table 6.2)	$M_{pl,Rd} = 0.25\cdot l_{eff}\cdot t_p^2\cdot f_y/\gamma_{M0}$; without prying $\min(2\cdot M_{pl,Rd}/m, n_t\cdot F_{t,Rd})$; with prying the modes $4\cdot M_{pl,Rd}/m$, $(2\cdot M_{pl,Rd} + n\cdot\Sigma F_{t,Rd})/(m + n)$ with $n = \min(e_{min}, 1.25m)$, and $n_t\cdot F_{t,Rd}$; l_{eff} is the total for the n_t bolts
Compression T-stub (cl. 6.2.6.9, 6.2.6.10)	$C_{Rd} = f_{jd}\cdot A_{flange}$, checked when $M_{Ed} > 0$ and the resultant is in compression
Shear (cl. 6.2.2)	$F_{2,vb,Rd} = \alpha_{bc}\cdot f_{ub}\cdot A_s/\gamma_{M2}$, $\alpha_{bc} = 0.44 − 0.0003\cdot f_{yb}$; $F_{vb,Rd} = \min(F_{1,vb,Rd}, F_{2,vb,Rd})$; $V_{Rd} = C_{f,d}\cdot N_{c,Ed} + n_b\cdot F_{vb,Rd}$; $V_{b,Ed} = \max(V_{Ed} − C_{f,d}\cdot N_{c,Ed}, 0)/n_b$
Required thickness	the smallest t_p in 10–100 mm with u_N, u_C and u_T all ≤ 1, z_C, C_{Ed} and T_{Ed} re-evaluated at every trial t_p

Basis: Design resistance of column bases in shear[1](cl. 6.2.2); Equivalent T-stub in tension[1](cl. 6.2.4, Table 6.2); Equivalent T-stub in compression[1](cl. 6.2.5); Concrete in compression including grout[1](cl. 6.2.6.9); Base plate in bending under compression[1](cl. 6.2.6.10); Base plate in bending under tension[1](cl. 6.2.6.11); Anchor bolts in tension[1](cl. 6.2.6.12); Column bases subject to axial forces[1](cl. 6.2.8.2); Column bases subject to axial force and bending moment[1](cl. 6.2.8.3, Table 6.7); Design strength of concrete; partial area loading[2](cl. 3.1.6, 6.7).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
column	Column section type	I/H		drawing: PF-GA-001 rev C	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
prying	Prying forces develop (short bolts, $L_b \le L_b^*$); off is conservative for the plate	no		assumption: Cast-in holding-down bolts: prying not assumed (min(2·M_pl/m, ΣF_t) is the conservative case).	any
B	Plate width	400	mm	drawing: PF-BP-001 rev B	150 … 1200 mm
L	Plate length	450	mm	drawing: PF-BP-001 rev B	150 … 1200 mm
t_p	Plate thickness	25	mm	drawing: PF-BP-001 rev B	10 … 60 mm
n_t	Holding-down bolts in each bolt row (one row each side)	2		drawing: PF-BP-001 rev B	1 … 6
n_b	Holding-down bolts in total (both sides)	4		drawing: PF-BP-001 rev B	2 … 12
x_b	Bolt row distance from the column axis (outside the column face, inside the plate edge)	185	mm	drawing: PF-BP-001 rev B	80 … 600 mm
m	Bolt to flange-weld toe distance, tension T-stub (required: it follows from x_b, the column and the weld)	46.9	mm	assumption: Bolt at 185 mm from the axis, flange face at 130.15 mm: 54.85 mm, less 0.8·√2 × 7 mm (a 10 mm fillet leg has a 7 mm throat) = 7.9 mm.	10 … 100 mm
l_{eff}	Effective length of the tension T-stub, total for the n_t bolts (required: from the cl. 6.2.6.5 patterns for the plate)	350	mm	assumption: Total for the two bolts of the row, about 175 mm each from the cl. 6.2.6.5 patterns (engineer's estimate).	50 … 800 mm
h	Column depth, along the plate length L (I/H and RHS)	260.3	mm	spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy	100 … 1000 mm
b	Column width, along the plate width B (I/H and RHS)	256.3	mm	spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy	100 … 1000 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
t_f	Flange thickness (I/H)	17.3	mm	spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy	5 … 50 mm
t_w	Web thickness (I/H)	10.3	mm	spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy	4 … 40 mm
t_g	Grout thickness (optional; blank = the β_j = 2/3 condition is not checked)	30	mm	drawing: PF-BP-001 rev B	5 … 100 mm
f_y	Plate yield strength	265	MPa	spec: BS EN 10025-2 S275, 16 mm < t ≤ 40 mm	235 … 440 MPa
f_{ck}	Concrete characteristic cylinder strength	30	MPa	spec: BS EN 206 C30/37 pad foundation, drawing PF-FN-001	20 … 45 MPa
$F_{t,Rd}$	Tension resistance per bolt (from the connection sheet)	203	kN	reference: S4 bolt sheet, M24 8.8 (203.04 kN), rounded down	10 … 1000 kN
f_{ub}	Anchor bolt ultimate tensile strength	800	MPa	spec: EN ISO 898-1 class 8.8	400 … 900 MPa
f_{yb}	Anchor bolt yield strength (α_{bc} is valid for 235–640 MPa)	640	MPa	spec: EN ISO 898-1 class 8.8	240 … 640 MPa
A_s	Anchor bolt tensile stress area (from the connection sheet)	352.5	mm²	reference: S4 bolt sheet, M24 stress area (352.50 mm²)	80 … 1000 mm²
$F_{1,vb,Rd}$	Bolt bearing resistance on the plate (from the bearing resistance of the connection sheet)	364.4	kN	reference: S4 bolt sheet, M24 bearing on 25 mm S275 plate, e_1 = 60 mm, d_o = 27 mm (364.44 kN)	10 … 1000 kN
$f_{ck,g}$	Grout characteristic strength (optional; blank = the β_j = 2/3 condition is not checked)	60	MPa	spec: Non-shrink grout datasheet, 28-day strength (placeholder GR-DS-01)	20 … 100 MPa
N_{Ed}	Axial force, compression positive and opposite in sign to the member sheet's axial force (a frame reaction $R1_y$ is compression positive)	210	kN	reference: S5 load combinations sheet, ULS 6.10b, wind leading, column base A	-2000 … 5000 kN

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
M_{Ed}	Major-axis moment (uniaxial)	95	kN·m	reference: S5 load combinations sheet / S2 frame sheet, column base A	0 … 1000 kN·m
V_{Ed}	Shear force	60	kN	reference: S5 load combinations sheet / S2 frame sheet, column base A	0 … 1000 kN
α_{cc}	Coefficient for long-term effects on the concrete strength	0.85		spec: UK NA to BS EN 1992-1-1, NA to cl. 3.1.6(1)P	0.8 … 0.9
γ_c	Partial factor for concrete	1.5		spec: UK NA to BS EN 1992-1-1, Table NA.1	1.2 … 1.6
γ_{M0}	Partial factor for the plate	1		spec: UK NA to BS EN 1993-1-1, NA.2.15	1 … 1.15
γ_{M2}	Partial factor for the anchor bolts	1.25		spec: UK NA to BS EN 1993-1-8, NA.2.3	1.1 … 1.4
C_{fd}	Friction coefficient, plate to grout	0.2		spec: EN 1993-1-8 cl. 6.2.2, sand-cement grout	0 … 0.4
α	Concentration factor $\sqrt{(A_{c1}/A_{c0})}$ of the foundation (1 when the foundation geometry is not known)	1.5		assumption: Conservative: a 1.2 m square pad under the 0.45 × 0.40 m plate gives A_c1/A_c0 well above 2.25; checked on drawing PF-FN-001.	1 … 2.5
α_{basis}	Foundation geometry that the concentration factor rests on (state it when the factor is above 1)	Pad 1.2 × 1.2 m, 0.8 m deep (PF-FN-001): A_c1/A_c0 above 2.25, so α = 1.5 is taken		drawing: PF-FN-001 rev A	any
$profis_{ref}$	Hilti PROFIS report reference (the report is filed as Evidence)	PROFIS-BP1 (to be filed)		assumption: The PROFIS report is not attached in this example, so the sheet warns that the anchorage is not checked.	any

3. Assumptions

A1. The axial force is positive in compression, the opposite of the member check sheet. (Sign convention.)
A2. A link from the frame sheet takes the reaction $R1_y$, never the member force N_{e1}. ($R1_y$ is up positive, so compression into the base.)
A3. Loading is static, with a major-axis moment only. (Scope of this sheet.)
A4. The plate is unstiffened and rectangular, on grout adequate for β_j = 2/3. (The grout is checked when its strength and thickness are entered.)
A5. The simplified T-stub model has one compression resultant and one tension resultant per bolt row. (BS EN 1993-1-8:2005 cl. 6.2.8.3.)
A6. The compression resultant acts at the flange centre, or at the half-annulus centroid of a CHS. (BS EN 1993-1-8:2005 cl. 6.2.8.3.)
A7. The force regime follows Table 6.7, and the web is not credited in the moment check. (BS EN 1993-1-8:2005 cl. 6.2.8.3.)
A8. The effective length l_{eff} is the total for the n_t bolts of a row. (Read from the cl. 6.2.6.5 patterns.)
A9. One l_{eff} serves modes 1 and 2, which is conservative for mode 2. ($l_{eff,1}$ is not more than $l_{eff,2}$.)
A10. The concrete, pull-out, breakout, edge and interaction failures of the anchors are designed in Hilti PROFIS. (From $F_{t,Ed}$ and $V_{b,Ed}$ per anchor; the report is filed as Evidence and referenced.)
A11. With prying switched on, the prying force Q is not in $F_{t,Ed}$ and PROFIS must add it. (External anchorage design.)
A12. The friction resistance counts only the compression, $C_{f,d}$·max(N_{Ed}, 0). (BS EN 1993-1-8:2005 cl. 6.2.2.)
A13. A column in net tension has no friction resistance. (BS EN 1993-1-8:2005 cl. 6.2.2.)
A14. Anchor bolts share the shear equally. (BS EN 1993-1-8:2005 cl. 6.2.2.)
A15. Anchor bolt holes are no larger than the standard clearance. (Oversized or slotted holes are outside this sheet.)
A16. The bearing outstand c is cut back where it would pass the plate edge. (Conservative against an uncut c.)
A17. The web and the column interior keep the full outstand c. (Conservative against an uncut c.)
A18. The CHS compression side is half the annular bearing area, with its centroid as lever arm. (An approximation.)
A19. The compression T-stub is checked against the flange area alone when M_{Ed} is above 0. (Conservative limitation.)
A20. The whole footprint resists at M_{Ed} = 0, so the compression check jumps just above 0, conservatively. (Limitation.)
A21. Partial factors and the friction coefficient come from the records in use. (γ_c and α_{cc}: EN 1992-1-1 NA; γ_{M2}: EN 1993-1-8 NA; γ_{M0}: EN 1993-1-1 NA; $C_{f,d}$: the grout used.)
A22. The defaults are the EN recommended and UK values. (A small set of factors only; no table from a standard is shipped.)
A23. prying (Prying forces develop (short bolts, $L_b \le L_b^*$); off is conservative for the plate) is taken as no. (Cast-in holding-down bolts: prying not assumed (min($2 \cdot M_{pl}/m$, ΣF_t) is the conservative case).)
A24. m (Bolt to flange-weld toe distance, tension T-stub (required: it follows from x_b, the column and the weld)) is taken as 46.9 mm. (Bolt at 185 mm from the axis, flange face at 130.15 mm: 54.85 mm, less 0.8·√2 × 7 mm (a 10 mm fillet leg has a 7 mm throat) = 7.9 mm.)
A25. l_{eff} (Effective length of the tension T-stub, total for the n_t bolts (required: from the cl. 6.2.6.5 patterns for the plate)) is taken as 350 mm. (Total for the two bolts of the row, about 175 mm each from the cl. 6.2.6.5 patterns (engineer's estimate).)
A26. α (Concentration factor $\sqrt{(A_{c1}/A_{c0})}$ of the foundation (1 when the foundation geometry is not known)) is taken as 1.5. (Conservative: a 1.2 m square pad under the 0.45 × 0.40 m plate gives A_{c1}/A_{c0} well above 2.25; checked on drawing PF-FN-001.)
A27. $profis_{ref}$ (Hilti PROFIS report reference (the report is filed as Evidence)) is taken as PROFIS-BP1 (to be filed). (The PROFIS report is not attached in this example, so the sheet warns that the anchorage is not

checked.)

4. Working

[31] $f_{cd} = \alpha_{cc} * f_{ck} / \gamma_c$ [2](cl. 3.1.6)
$= 0.85 \times 30\ \text{MPa} / 1.5$
$= 17\ \text{MPa}$
Concrete design compressive strength

[32] $\beta_j = 2/3$, fixed [1](cl. 6.2.5)
= grout conditions of cl. 6.2.5 assumed met
$= 0.66667$
Joint coefficient β_j

[33] $f_{jd} = \beta_j * \alpha * f_{cd}$ [1](cl. 6.2.5)
$= 0.66667 \times 1.5 \times 17\ \text{MPa}$
$= 17\ \text{MPa}$
Design bearing strength of the joint

[34] $c = t_p * \text{sqrt}(f_y / (3 * f_{jd} * \gamma_{M0}))$ [1](cl. 6.2.5)
$= 25\ \text{mm} \times \text{sqrt}(265\ \text{MPa} / (3 \times 17\ \text{MPa} \times 1))$
$= 56.987\ \text{mm}$
Additional bearing width round the column footprint

[35] $A_{eff} = 2(b + 2c)(t_f + 2c) + (h - 2t_f - 2c)(t_w + 2c)$ [1](cl. 6.2.5)
$= 2 \times 370.27 \times 131.27 + 111.73 \times 124.27\ \text{mm}^2$
$= 111100\ \text{mm}^2$
Effective bearing area

[38] $N_{j,Rd} = f_{jd} * A_{eff}$ [1](cl. 6.2.8.2, 6.2.5)
$= 17\ \text{MPa} \times 111100\ \text{mm}^2$
$= 1888.7\ \text{kN}$
Axial bearing resistance of the joint

[39] $N_{c,Ed} = \max(N_{Ed}, 0)$
$= N_{Ed} = 210\ \text{kN}$
$= 210\ \text{kN}$
Design compression in the column

[40] $u_N = N_{c,Ed} / N_{j,Rd}$
$= 210\ \text{kN} / 1888.7\ \text{kN}$
$= 0.11119$
Axial bearing utilisation

[41] $t_{req,N}$ = smallest t_p in 10–100 mm with $f_{jd} \cdot A_{eff}(c(t_p)) \geq N_{c,Ed}$ (bisection) [1](cl. 6.2.5)
$= N_{c,Ed} = 210\ \text{kN}$; carried at the 10 mm minimum
$= 10\ \text{mm}$
Plate thickness required for axial bearing alone

[42] $z_C = (h - t_f)/2$ [1](cl. 6.2.8.3)
$= (260.3 - 17.3)/2\ \text{mm}$
$= 121.5\ \text{mm}$
Compression resultant distance from the column axis

[43] $z = z_C + x_b$ [1](cl. 6.2.8.3)
$= 121.5 + 185$
$= 306.5\ \text{mm}$
Lever arm between the compression resultant and the bolt row

[44] $C_{Ed} = (M_{Ed} + N_{Ed} \cdot x_b)/z$ [1](cl. 6.2.8.3, Table 6.7)
$= (95\ \text{kN·m} + 210\ \text{kN} \times 185\ \text{mm})/306.5\ \text{mm}$
$= 436.7\ \text{kN}$
Compression resultant under the compression flange

[45] $T_{Ed} = C_{Ed} - N_{Ed}$ [1](cl. 6.2.8.3, Table 6.7)
$= 436.7 - 210$
$= 226.7\ \text{kN}$
Tension resultant at the bolt row

[46] $C_{Ed,2}$ = none
= other side
= 0 kN
Compression resultant under the other flange
[47] $T_{Ed,2}$ = none
= other row
= 0 kN
Tension resultant at the other bolt row
[48] $F_{t,Ed} = T_{Ed} / n_t$ [1](cl. 6.2.6.12)
= 226.7 kN / 2
= 113.35 kN
Holding-down bolt tension per bolt at the more loaded row (→ PROFIS)
[49] A_{flange} = (b + 2c)(t_f + 2c), at most $A_{eff}/2$ [1](cl. 6.2.6.9)
= min(370.27 × 131.27, $A_{eff}/2$) mm²
= 48608 mm²
Bearing area of the compression flange
[50] $C_{Rd} = f_{jd} * A_{flange}$ [1](cl. 6.2.6.9, 6.2.6.10)
= 17 MPa × 48608 mm²
= 826.33 kN
Bearing resistance of the compression flange
[51] $u_C = C_{Ed} / C_{Rd}$
= 436.7 kN / 826.33 kN
= 0.52849
Compression flange utilisation
[52] $M_{pl,Rd} = 0.25 * l_{eff} * t_p$^2 * f_y / γ_{M0} [1](Table 6.2)
= 0.25 × 350 mm × (25 mm)^2 × 265 MPa / 1
= 14.492 kN·m
Plastic moment resistance of the plate over l_{eff}
[53] $T_{Rd,1} = 2 * M_{pl,Rd} / m$ [1](Table 6.2)
= 2 × 14.492 kN·m / 46.9 mm
= 618 kN
Mode 1-2 without prying forces
[54] $T_{Rd,3} = n_t * F_{t,Rd}$ [1](Table 6.2)
= 2 × 203 kN
= 406 kN
Mode 3: bolt failure
[55] $T_{Rd} = \min(T_{Rd,1}, T_{Rd,3})$ [1](cl. 6.2.4)
= min(618 kN, 406 kN)
= 406 kN
Tension T-stub resistance (no prying)
[56] $u_T = T_{Ed} / T_{Rd}$
= 226.7 kN / 406 kN
= 0.55839
Plate bending and bolts in tension, T-stub utilisation
[57] $u_{bt} = F_{t,Ed} / F_{t,Rd}$
= 113.35 kN / 203 kN
= 0.55839
Holding-down bolt tension utilisation
[58] t_{req} = smallest t_p in 10–100 mm with u_N, u_C, $u_T \le 1$, with z_C, C_{Ed} and T_{Ed} recomputed at every trial t_p (scan and bisection) [1](cl. 6.2.5, 6.2.6.9, 6.2.6.11)
= $N_{c,Ed}$ = 210 kN, C_{Ed} = 436.7 kN, T_{Ed} = 226.7 kN
= 15.142 mm
Plate thickness required

[59] $\alpha_{bc} = 0.44 - 0.0003 \cdot f_{yb}$ [1](cl. 6.2.2)
$= 0.44 - 0.0003 \times 640$
$= 0.248$
Anchor bolt shear factor α_{bc}
[60] $F_{2,vb,Rd} = \alpha_{bc} * f_{ub} * A_s / \gamma_{M2}$ [1](cl. 6.2.2)
$= 0.248 \times 800\ MPa \times 352.5\ mm^2 / 1.25$
$= 55.949\ kN$
Anchor bolt shear resistance $F_{2,vb,Rd}$
[61] $F_{vb,Rd} = \min(F_{1,vb,Rd}, F_{2,vb,Rd})$ [1](cl. 6.2.2)
$= \min(364.4\ kN, 55.949\ kN)$
$= 55.949\ kN$
Anchor bolt shear resistance per bolt (bearing or shear)
[62] $F_{f,Rd} = C_{fd} * N_{c,Ed}$ [1](cl. 6.2.2)
$= 0.2 \times 210\ kN$
$= 42\ kN$
Friction resistance, compression only
[63] $V_{Rd} = F_{f,Rd} + n_b * F_{vb,Rd}$ [1](cl. 6.2.2)
$= 42\ kN + 4 \times 55.949\ kN$
$= 265.8\ kN$
Shear resistance of the base
[64] $u_V = V_{Ed} / V_{Rd}$
$= 60\ kN / 265.8\ kN$
$= 0.22574$
Shear utilisation
[65] $V_{b,Ed} = \max(V_{Ed} - F_{f,Rd}, 0)/n_b$ [1](cl. 6.2.2)
$= \max(60 - 42, 0)/4$
$= 4.5\ kN$
Shear per anchor bolt after friction (→ PROFIS)
[66] $u_{max} = \max(u_N, u_C, u_T, u_{bt}, u_V)$
$= u_N = 0.11119,\ u_C = 0.52849,\ u_T = 0.55839,\ u_{bt} = 0.55839,\ u_V = 0.22574$
$= 0.55839$
Governing utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
$F_{t,Ed}$	113.35 kN	Bolt tension per bolt (to PROFIS)
$V_{b,Ed}$	4.500 kN	Shear per anchor bolt (to PROFIS)
$N_{j,Rd}$	1888.7 kN	Axial bearing resistance
u_{max}	0.56	Governing utilisation
f_{cd}	17 MPa	Concrete design strength
β_j	0.66667	Joint coefficient β_j
f_{jd}	17 MPa	Design bearing strength of the joint
c	56.99 mm	Bearing outstand round the column footprint
A_{eff}	111100 mm²	Effective bearing area
$t_{req,N}$	10 mm	Plate thickness required for axial bearing alone
t_{req}	15.14 mm	Plate thickness required (bearing, compression and tension T-stubs)

SYMBOL	VALUE	DESCRIPTION
z_C	121.5 mm	Distance from the column axis to the compression resultant
z	306.5 mm	Lever arm between the compression resultant and the bolt row
C_{Ed}	436.70 kN	Compression resultant
$C_{Ed,2}$	0 kN	Compression resultant, other side
T_{Ed}	226.70 kN	Tension resultant, more loaded row (governing)
$T_{Ed,2}$	0 kN	Tension resultant, other row
A_{flange}	48608 mm²	Compression-side bearing area
C_{Rd}	826.33 kN	Compression-side resistance
$M_{pl,Rd}$	14.49 kN·m	Plate plastic moment
$T_{Rd,1}$	618.00 kN	T-stub mode 1-2, no prying
$T_{Rd,3}$	406 kN	T-stub mode 3
T_{Rd}	406 kN	Tension T-stub resistance
α_{bc}	0.248	Anchor shear factor α_{bc}
$F_{2,vb,Rd}$	55.95 kN	Anchor shear resistance $F_{2,vb,Rd}$
$F_{vb,Rd}$	55.95 kN	Anchor shear resistance per bolt
$F_{f,Rd}$	42 kN	Friction resistance
V_{Rd}	265.80 kN	Shear resistance
regime	"tension at one bolt row, compression under the other flange (Table 6.7, z = z_C + x_b)"	Force regime (Table 6.7)
T_{mode}	"mode 3 (bolt failure)"	Governing T-stub mode
u_N	0.11	Axial bearing utilisation
u_C	0.53	Compression-side utilisation
u_T	0.56	Tension T-stub utilisation
u_{bt}	0.56	Bolt tension utilisation
u_V	0.23	Shear utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Tension at the bolt row against the plate T-stub resistance	226.70 kN	406 kN	0.56	[1](cl. 6.2.6.11)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Bolt tension against the bolt tension resistance	113.35 kN	203 kN	0.56	[1](cl. 6.2.6.12)
PASS	Compression force against compression flange bearing resistance	436.70 kN	826.33 kN	0.53	[1](cl. 6.2.6.9, 6.2.6.10)
PASS	Shear force against friction plus anchor bolt shear resistance	60 kN	265.80 kN	0.23	[1](cl. 6.2.2)
PASS	Axial force against concrete bearing resistance	210 kN	1888.7 kN	0.11	[1](cl. 6.2.8.2, 6.2.5)

7. Applicability

WARNING records: tabulated values not taken from a reference record are α_{cc}, γ_c (EN 1992-1-1 National Annex factors); γ_{M2} (EN 1993-1-8 National Annex factor); γ_{M0} (EN 1993-1-1 National Annex factor); C_{fd} (Friction coefficient plate to grout); f_y (Plate strength). Enter each as one so its source, edition and licence travel with it. [record]
WARNING f_y: one strength of 265 MPa is used at every thickness, but EN 10025-2 lowers it above 16 mm. Enter the strength for the thickness in use, because t_p exceeds that step. [$f_{y,thickness}$]
WARNING u_{profis}: the anchorage is not assessed here, and u_{max} covers the steel parts only. Enter the PROFIS report's governing utilisation to include the concrete failure modes of the anchors. [anchorage-result]
WARNING $profis_{ref}$: 'PROFIS-BP1 (to be filed)' is typed text, not an attached report. Attach the PROFIS report as Evidence, because the anchorage is not checked here. [anchorage]

8. Verification

```
Template struct.base-plate@1.7.0 Content hash b3f3c18d0ee3f47a Engine calc-engine/1.0.0
Known-answer cases passed 9/9 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-8: Design of Joints. BS EN 1993-1-8:2005. BSI; 2005.
2. BS EN 1992-1-1:2004.